



Property PRS Pricing

User Guide — Property-Level Physical Risk Swap Valuation

MKM Research Labs

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1 Introduction

A **Property PRS** extends the Physical Risk Swap concept from a river gauge to a specific property. Instead of triggering on a gauge water level, the contract references a named property and its flood exposure.

1.1 Why Property-Level Pricing Matters

A gauge measures water level at a single point on the river. Two properties at different distances and elevations relative to that gauge experience very different flood depths from the same event. A property on high ground 2 km from the Thames may never flood even when the gauge breaches severe, while a basement flat 200 m from the channel floods regularly.

Property PRS captures this distinction. The fair spread reflects the *property-specific* flood probability, not the gauge-level probability. This is always lower than or equal to the gauge spread, because the gauge is the most exposed point on the river.

1.2 Gauge PRS vs. Property PRS

	Gauge PRS	Property PRS
Reference entity	River gauge	Named property
Pricing method	Analytical (hazard rate)	Event-count (Monte Carlo)
Term structure	Upward-sloping	Flat (storms independent)
Basis to gauge	Zero (is the gauge)	Positive (property safer)

2 Event-Count Pricing

Property PRS spreads are derived from the Monte Carlo storm catalogue rather than analytical hazard rates.

2.1 The Formula

$$s_{\text{bps}} = \frac{\text{severe floods at property}}{\text{total scenarios}} \times 10,000$$

For example, if 45 out of 1,000 storm scenarios produce severe flooding at a property, the fair spread is:

$$s_{\text{bps}} = \frac{45}{1,000} \times 10,000 = 450 \text{ bps}$$

2.2 Why the Term Structure Is Flat

Each storm scenario is an independent 168-hour event. The probability of a severe flood in any given year does not depend on the contract's remaining maturity. Therefore the property PRS spread is the same at all tenors (1Y, 2Y, 3Y, 5Y).

This differs from gauge PRS, where the analytical survival function $S(t) = e^{-\lambda t}$ naturally produces an upward-sloping term structure (longer tenors have higher cumulative default probability).

2.3 What Counts as a “Severe Flood”

A storm scenario produces a severe flood at a property when:

1. The nearest synthetic gauge’s water level exceeds the severe warning threshold.
2. The flood signal propagates to the property (accounting for distance, retention, elevation, and infiltration).
3. The resulting flood depth at the property exceeds zero.

3 Basis Spread Decomposition

The **basis** is the difference between the gauge spread and the property spread:

$$\text{basis}_{\text{bps}} = s_{\text{gauge}} - s_{\text{property}}$$

The basis is always non-negative because the property is never more exposed than its controlling gauge.

3.1 What Drives the Basis

Two physical factors create the basis:

Spatial Heterogeneity Effect (SHE) Not every gauge-level severe event produces flooding at the property. The spatial correlation model distributes rainfall unevenly, so some storms hit the gauge hard but miss the property’s nearest synthetic gauge.

Surface Height Distance Effect (SHD) Even when the synthetic gauge floods, the property may be elevated or far enough from the channel that the flood depth at the property is zero. Distance retention, terrain velocity, and floor level all attenuate the signal.

3.2 Transmission Rate

The **transmission rate** quantifies how efficiently gauge floods translate to property floods:

$$T = \frac{\text{severe floods at property from this gauge}}{\text{severe events at gauge}}$$

A transmission rate of 0.90 means 90% of gauge-level severe events also produce flooding at the property. A rate of 0.30 means only 30% transmit — the basis is large and the property is well protected.

3.3 IDW Reconstruction

When a property has multiple nearest gauges, the property spread is reconstructed from the gauge spreads minus basis via inverse-distance weighting (IDW):

$$s_{\text{property}} = \sum_g w_g \times (s_g - b_g)$$

where $w_g = 1/d_g$ normalised to sum to 1, and b_g is the basis for gauge g . In practice, the nearest synthetic gauge dominates.

4 Property Trade Structure

A Property PRS trade extends the gauge PRS CDM format with a **PropertySet** block:

Field	Description
PropertyID	Unique identifier (e.g. PROP-00012345)
PropertyAddress	Street address
Postcode	UK postcode
LocalAuthority	Council area
PropertyValue	Market value (£)
EAFloodZone	EA flood zone (Zone 3a, 2, 1)
ReferenceGauge	Nearest synthetic gauge ID
Latitude / Longitude	WGS84 coordinates

The trade also carries a **GaugeBasket** with the controlling gauge(s) and their weights, plus a **basis_bps** field recording the gauge-to-property spread difference at inception.

5 Property-Level Greeks

5.1 Basis Delta

The property FS01 is driven by the gauge hazard rate. When the gauge hazard rate moves by 1 bp, the gauge spread moves by Δs bps. The property spread moves by the same amount (the basis is held constant):

$$\text{FS01}_{\text{property}} = \frac{\Delta s}{10,000} \times A \times N$$

This is identical to the gauge FS01 because the basis is a structural constant determined by the property's physical position, not by market movements.

5.2 Basis Risk

The basis does change when the storm catalogue is regenerated (e.g. with different storm control parameters). This represents *model risk*, not market risk. It is not captured in FS01 but is visible in stress testing.

6 Parameter Reference

Parameter	Default	Increase	Decrease
N_NEAREST_GAUGES	3	More gauges contribute via IDW	Nearest gauge dominates
BANKFULL_OFFSET_M	0.5 m	Property floods less often (wider basis)	Property floods more often
RETENTION_LENGTH	10,000 m	Flood signal reaches further (narrower basis)	Signal attenuates faster (wider basis)
EA_FLOOD_ZONE_RATES	0.1–5%	Higher zone-based pricing	Lower zone-based pricing
DEPTH_POINTS	see text	Stretches damage curve	Compresses damage curve
DAMAGE_POINTS	see text	More damage per depth	Less damage per depth
DEFAULT_IMPERV	0.4	Less infiltration, deeper floods	More infiltration, shallower floods

Document History

Version	Date	Author	Description
1.0	10-Apr-2026	MKM Research Labs	Initial release